it suffices by Lemma 8.6.7 to show that each of them has an edge in every finite cut F of G. Given F, choose n large enough that all the edges of F lie in $G[v_0, \dots, v_n]$. Then F is also a cut of G_n. Now consider the k-tuple $(H_n^1, \dots, H_n^k)$ which the infinity lemma picked from $\mathcal{V}_n$. Each of these H_n^i is a connected spanning submultigraph of G_n, so it contains an edge from F. But H_n^i agrees with H^i on $\{v_0, \dots, v_n\}$, so H^i too contains this edge from F. $\square$

Lemma 8.6.11. *Every connected standard subspace of $|G|$ that contains V also contains a topological spanning tree of G.*

Proof. Let X be a connected standard subspace of $|G|$ containing V. Then G too must be connected, so it is countable. Let $e_0, e_1, \dots$ be an enumeration of $E(X)$, and consider these edges in turn. Starting with $X_0 := X$, define $X_{n+1} := X_n \smallsetminus \mathring{e}_n$ if this keeps X_{n+1} connected; if not, put $X_{n+1} := X_n$. Finally, let $T := \bigcap_{n \in \mathbb{N}} X_n$.

Since T is closed and contains V, it is still a standard subspace. And T has an edge in every finite cut of G, because X does and its last edge in that cut will never be deleted. So T is connected, by Lemma 8.6.7. But T contains no circle: that would contain an edge, which should have got deleted since deleting an edge from a circle cannot destroy connectedness. $\square$

Proof of Theorem 8.6.9. The implication (ii)→(i) follows from our two lemmas. For (i)→(ii), let G have edge-disjoint topological spanning trees $T_1, \dots, T_k$, and consider a partition P of V into ℓ sets. If there are infinitely many cross-edges, there is nothing to show; so we assume there are only finitely many. For each $i \in \{1, \dots, k\}$, let T_i' be the multigraph of order ℓ which the edges of T_i induce on P.

To establish that G has at least $k(\ell - 1)$ cross-edges, we show that the multigraphs T_i' are connected. If not, then some T_i' has a vertex partition crossed by no edge of T_i. This partition induces a cut of G that contains no edge of T_i. By our assumption that G has only finitely many cross-edges, this cut is finite. By Lemma 8.6.7, this contradicts the connectedness of T_i. $\square$

(1.9) 8.7 The topological cycle space

As a more comprehensive application of our new theory, let us now look at how the cycle space theory of finite graphs extends to locally finite
$G = (V, E)$ graphs $G = (V, E)$ with infinite circuits and topological spanning trees.

Every two points of a topological spanning tree T are joined by a unique arc in T: existence follows from Lemma 8.6.5, while uniqueness is proved as for finite graphs. Adding a new edge e to T therefore creates a unique circle in $T \cup e$; its edges form the *fundamental circuit* C_e of e with respect to T. Note that C_e can be infinite. *fundamental circuit C_e*

Similarly, for every edge $f \in E(T)$ the space $T \setminus \mathring{f}$ has exactly two arc-components; the set of edges between these is the *fundamental cut* D_f of T. Since the two arc-components of $T \setminus \mathring{f}$ are closed (Lemma 8.6.6) but disjoint, Lemma 8.6.3 (ii) implies that D_f is finite. *fundamental cut D_f*

As in finite graphs, we have $e \in D_f$ if and only if $f \in C_e$, for all $f \in E(T)$ and $e \in E \setminus E(T)$. Topological spanning trees that are the closure of a normal spanning tree, as in Lemma 8.6.8, are particularly useful in this context: their fundamental circuits and cuts are both finite. (8.6.8)

For locally finite graphs there will be two cycle spaces: the usual 'finitary' one from Chapter 1.9, and a new 'topological' one based on topological circuits. The former will be a subspace of the latter, much as the space of all finite cuts is a subspace of the space of all cuts. These four spaces are cross-related by matroid duality in a surprising way; see the notes and Exercise 123.

Call a family $(D_i)_{i \in I}$ of subsets of E *thin* if no edge lies in D_i for infinitely many i. Let the *thin sum* $\sum_{i \in I} D_i$ of this family be the set of all edges that lie in D_i for an odd number of indices i. The *topological cycle space* $\mathcal{C}(G)$ of G is the subspace of its edge space $\mathcal{E}(G)$ consisting of all thin sums of circuits. *thin sum* *topological cycle space $\mathcal{C}(G)$*

We say that a given set $\mathcal{Z}$ of circuits *generates* $\mathcal{C}(G)$ if every element of $\mathcal{C}(G)$ is a thin sum of elements of $\mathcal{Z}$. For example, the topological cycle space of the ladder in Figure 8.1.3 can be generated by all its squares (the 4-element circuits), or by the infinite circuit consisting of all horizontal edges and all squares but one. Similarly, the 'wild circuit' of Figure 8.6.1 is the thin sum of all the finite face boundaries of that graph, which thus generate it. *generates*

Let us use $\mathcal{C}_{\text{fin}}(G)$ to denote the *finitary cycle space* of G as defined in Chapter 1.9: the (finite) sums of its finite circuits. Clearly $\mathcal{C}_{\text{fin}}(G) \subseteq \mathcal{C}(G)$. We shall see later that $\mathcal{C}_{\text{fin}}(G)$ contains all the finite elements of $\mathcal{C}(G)$, but this is not obvious from the definitions; see Exercise 120. When G is finite, however, clearly $\mathcal{C}_{\text{fin}}(G) = \mathcal{C}(G)$. *finitary cycle space $\mathcal{C}_{\text{fin}}(G)$*

As shown in Chapter 1.9, a finite set of edges of G lies in $\mathcal{C}_{\text{fin}}(G)$ if and only if it meets every cut of G evenly, and the fundamental circuits of any ordinary spanning tree generate $\mathcal{C}_{\text{fin}}(G)$ by finite sums: just copy the proofs given there. For $\mathcal{C}(G)$ we have the following topological analogue:

Theorem 8.7.1. *The following statements are equivalent for every set D of edges of a locally finite connected graph G:*

(i) *$D \in \mathcal{C}(G)$;*

(ii) *D meets every finite cut F of G in an even number of edges;*

(iii) *D is a thin sum of fundamental circuits of any topological spanning tree of G.*

(8.2.4)
(8.6.3)
(8.6.7)

Proof. The implication (iii)→(i) holds by definition of $\mathcal{C}(G)$ and the fact that G has a topological spanning tree (Lemma 8.6.11).

Let us prove (i)→(ii). By assumption, D is a thin sum of circuits. Only finitely many of these can meet F, so it suffices to show that every circuit meets F evenly. This follows from Lemma 8.6.3 (i): given a circle C in $|G|$, the segments of C between its edges in F (if any) are arcs whose vertices all lie on the same side of the cut F. These sides alternate as we follow C round. Therefore, there is an even number of such arcs, and hence also of edges that C has in F.

It remains to prove (ii)→(iii). Write C_e for the fundamental circuit of an edge $e \notin E(T)$, and D_f for the fundamental cut of an edge $f \in E(T)$. Recall that, by Lemma 8.6.3 (ii), these D_f are finite cuts. We show that

$$D = \sum_{e \in D \smallsetminus E(T)} C_e \,. \qquad (*)$$

This sum is well defined: since $f \in C_e \Leftrightarrow e \in D_f$ and fundamental cuts are finite, the C_e in this sum form a thin family. To prove $(*)$ we show that $D' := D + \sum_{e \in D \smallsetminus E(T)} C_e = \emptyset$.

Note first that $D' \subseteq E(T)$: any chord of T that lies in D also lies in exactly one of the C_e in the sum. Hence any $f \in D'$ is the unique edge of T, and hence of D', in the finite cut D_f, giving $|D' \cap D_f| = 1$. This is a contradiction, since D meets D_f evenly by (ii), and every C_e does by Lemma 8.6.3. □

Corollary 8.7.2. *$\mathcal{C}(G)$ is generated by finite circuits.*

(8.2.4) *Proof.* Apply Theorem 8.7.1 with the closure of a normal spanning tree, which is a topological spanning tree by Lemma 8.6.8. □

Our second aim in this section is to prove the analogue of Proposition 1.9.1 (ii) for the topological cycle space: that its elements D are not only thin sums but even disjoint unions of circuits. For finite graphs, it was easy to find these circuits greedily: we would 'follow the edges of D round' until a circuit was found, delete it, and repeat.

This will still be our overall strategy when G is infinite. But it is no longer straightforward now to isolate a single circuit from D. For example, without using our knowledge that the edge set D of the wild circle in the graph G of Figure 8.6.1 is a circuit, we can see at once that it must lie in $\mathcal{C}(G)$: it is the thin sum of all the finite circuits bounding a face. Our proof must therefore be able to 'decompose' D into disjoint circuits. Since D itself is the only circuit contained in D, the proof thus has to reconstruct the complicated wild circle just from the information that $D \in \mathcal{C}(G)$. And it has to do so generically, without appealing to the special structure of this particular graph.

Theorem 8.7.3. *For every locally finite graph G, every element of $\mathcal{C}(G)$ is a disjoint union of circuits.*

(1.9.1)
(8.1.2)
(8.6.4)

Proof. We may assume that G is connected, and hence countable. Let $D \in \mathcal{C}(G)$ be given, and enumerate its edges. We inductively construct a sequence of disjoint circuits $C \subseteq D$ each containing the smallest edge in our enumeration of D that is not yet contained in the circuits constructed before. Then all these circuits will form the desired partition of D.

Suppose we have already constructed finitely many disjoint circuits all contained in D. Deleting these edges from D leaves a set D' of edges that is again in $\mathcal{C}(G)$; let e be its smallest edge in our enumeration of D. We shall find a topological path π between the endvertices of e in the standard subspace that $D' \smallsetminus \{e\}$ spans in $|G|$. By Lemma 8.6.4, the image of π will contain an arc A between these vertices, and $A \cup e$ will be the circle defining our next circuit. D', e

Enumerate the vertices of G as $v_0, v_1, \ldots$, with $e = v_0v_1$. Let $S_n := \{v_0, \ldots, v_n\}$. For each $n \geqslant 1$, let G_n be the finite multigraph obtained from G by contracting every component of $G - S_n$ to a vertex, deleting any loops but keeping parallel edges that arise in the contraction. Note that both $V(G_n)$ and $E(G_n)$ are finite, and that $G[S_n] \subseteq G_n$. Let v_n' denote the vertex of $G_{n-1} - S_{n-1}$ whose branch set V_n contains v_n. $e = v_0v_1$ G_n v_n', V_n

We may think of $E(G_n)$ as a subset of $E(G)$. Then the cuts of G_n are also cuts of G. By Theorem 8.7.1, D' meets these evenly; in particular, every vertex of G_n is incident with an even number of edges in D'. Hence $D' \cap E(G_n) \in \mathcal{C}(G_n)$, by Proposition 1.9.1, so G_n contains a cycle through e that has all its edges in D'. Let P_n be the unique v_0–v_1 walk in this cycle that does not contain e and does not repeat any vertices.

Let $\mathcal{V}_n$ be the set of all v_0–v_1 walks in $G_n - e$ in which none of the vertices $v_0, \ldots, v_n$, and hence no edge, occurs more than once. Then $P_n \in \mathcal{V}_n \neq \emptyset$, and $\mathcal{V}_n$ is finite. Every walk $W \in \mathcal{V}_n$ with $n \geqslant 2$ induces a walk $W' \in \mathcal{V}_{n-1}$ consisting of the edges that W has in G_{n-1}, traversed in the same order and direction.[14] Thus, W' arises from W by replacing any subwalk of vertices and edges not in G_{n-1} with v_n'. The vertices of any such subwalk of W will be v_n or vertices of $G_n - S_n$ whose branch set is contained in V_n. By the infinity lemma, there exists a choice of walks $W_n \in \mathcal{V}_n$ such that $W_n' = W_{n-1}$ for all $n \geqslant 2$.

Our next aim is to turn these walks W_n into topological paths $\pi_n \colon [0,1] \to |G_n|$ that traverse them from v_0 to v_1 and reflect their compatibility. We shall define these π_n for $n = 1, 2, \ldots$ in turn, as follows.

For $n = 1$, note that W_1 has exactly two edges: at least two, because e has no parallel edge, and at most two, because every edge of G_1

[14] These are well defined: every edge $e \in W$ that is an edge of G_{n-1} has at least one endvertex in S_{n-1}, which either precedes it in W or follows it. In W', this vertex will likewise precede or follow e, respectively.

is adjacent to either v_0 or v_1. Let π_1 map $[0, \frac{1}{3}]$ onto the first edge, $[\frac{1}{3}, \frac{2}{3}]$ to the unique inner vertex of W_1, and $[\frac{2}{3}, 1]$ onto the second edge.

For $n \geqslant 2$, assume inductively that π_{n-1} traverses the edges of W_{n-1} in their given order and direction, and that π_{n-1} 'pauses' at each vertex $v \in G_{n-1} - S_{n-1}$ on W_{n-1} for a non-singleton closed interval $I \subseteq [0,1]$, mapping I constantly to that vertex. (Thus, if W_{n-1} visits v five times, then $\pi_{n-1}^{-1}(v)$ is a disjoint union of five such intervals.) We start our definition of π_n by letting $\pi_n(\lambda) := \pi_{n-1}(\lambda)$ for all λ with $\pi_{n-1}(\lambda) \in |G_n|$.

Every other $\lambda \in [0,1]$ satisfies $\pi_{n-1}(\lambda) = v'_n$. These λ form a disjoint union of closed intervals, one for every occurrence of v'_n on W_{n-1}. Recall that W_n arises from W_{n-1} by replacing each occurrence of v'_n by a subwalk of W_n whose vertices are either v_n or vertices of $G_n - S_n$ whose branch set is contained in V_n. For every occurrence of v'_n on W_{n-1}, let π_n on the corresponding interval I with $\pi_{n-1}(I) = \{v'_n\}$ traverse this subwalk of W_n, once more pausing for a non-singleton interval at any vertex that this subwalk has in $G_n - S_n$.

These maps π_n tend to a limit $\pi\colon [0,1] \to |G|$, defined as follows. Let $\lambda \in [0,1]$ be given. If $\pi_n(\lambda) \in |G|$ for some n, then $\pi_m(\lambda) = \pi_n(\lambda)$ for all $m > n$, and we let $\pi(\lambda) := \pi_n(\lambda)$. Otherwise $\pi_n(\lambda) \in V(G_n) \smallsetminus S_n$ for all n; let U_n be the branch set of this vertex $u_n := \pi_n(\lambda)$ of G_n in G. By our inductive construction of the maps π_n, we have $U_1 \supseteq U_2 \supseteq \ldots$. Since U_n spans a component $C_n = C_n(\lambda)$ of $G - S_n$, we can find a ray in G that has a tail in each C_n; let $\pi(\lambda)$ be the end ω of this ray. Note that ω, and hence $\pi(\lambda)$, is well defined: every end $\omega' \neq \omega$ is separated from ω by some S_n, and then fails to have a ray in C_n.

For a proof that π is our desired topological v_0–v_1 path in $|G|$, we need to check continuity at every λ. If $\pi(\lambda) = \pi_n(\lambda)$ for some n, then π agrees with π_n also in a small neighbourhood of λ, so this follows from the continuity of π_n. Otherwise $\pi(\lambda)$ is an end, ω say. Then ω has a neighbourhood basis in $|G|$ consisting of open sets $\hat{C}_\epsilon(S_n, \omega)$. Here $C(S_n, \omega)$ is the component $C_n(\lambda)$ defined earlier, since ω has a ray in it.

Now λ is an inner point of an interval $I \subseteq [0,1]$ which π_n maps to the vertex $u_n = \pi_n(\lambda)$. By construction, $\pi(I) \subseteq \overline{C_n(\lambda)} \subseteq \hat{C}_\epsilon(S_n, \omega)$, completing our continuity proof for π. □

Note that in the special case of Theorem 8.7.3 where the cycle space element considered is the entire edge set of G, Theorem 8.4.2 gives the stronger result that $E(G)$ is a disjoint union of finite circuits.

Corollary 8.7.4. *$\mathcal{C}(G)$ is closed under infinite thin sums.*

Proof. Consider a thin sum $\sum_{i \in I} D_i$ of elements of $\mathcal{C}(G)$. By Theorem 8.7.3, each D_i is a disjoint union of circuits. Together, these form a thin family, whose sum lies in $\mathcal{C}(G)$ and equals $\sum_{i \in I} D_i$. □

8.8 Infinite graphs as limits of finite ones

In the last section we saw how the space $|G|$, for a locally finite graph G, seems to appear as a 'limit' of the finite minors G_n of G obtained by contracting the components left on deleting its first n vertices. We now make this relationship between $|G|$ and the G_n more formal. Clarifying this can help a lot with transferring theorems for finite graphs to infinite ones – which, after all, is the idea behind considering $|G|$ in the first place.

directed *cofinal*

Let $(P, \leqslant)$ be a *directed* partially ordered set, one such that for all p, q there exists an r such that $p \leqslant r$ and $q \leqslant r$. A subset $Q \subseteq P$ is *cofinal* in P if for every $p \in P$ there exists some $q \in Q$ with $p \leqslant q$.

inverse system *bonding maps*

For every $p \in P$ let X_p be a compact Hausdorff topological space; later, these will represent finite graphs. Assume that we have continuous maps $f_{qp}\colon X_q \to X_p$ for all $q > p$, which are *compatible* in that, whenever $r > q > p$, we have $f_{qp} \circ f_{rq} = f_{rp}$. The family $\mathcal{X} = (\, X_p \mid p \in P\,)$, together with these *bonding maps* f_{qp}, is called an *inverse system*.

inverse limit

The set X of all $x = (\, x_p \mid p \in P\,)$ with $x_p \in X_p$ and $f_{qp}(x_q) = x_p$ for all $p < q$ in P is the *inverse limit* $X = \varprojlim \mathcal{X}$ of $\mathcal{X}$. We give it the subspace topology from the product space $\prod_{p \in P} X_p$ which, like the X_p, is Hausdorff and compact by Tychonoff's theorem.

The space $X = \varprojlim \mathcal{X}$ is the intersection, over all $q \in P$, of the sets $X_{<q}$ of all $(\, x_p \mid p \in P\,) \in \prod_p X_p$ that satisfy $f_{qp}(x_q) = x_p$ for all $p < q$. Using the fact that the X_p are Hausdorff and the maps f_{qp} are continuous, one can show that these subsets $X_{<q}$ of $\prod_p X_p$ are closed. Thus, $X = \bigcap_{q \in P} X_{<q}$ is closed in the compact space $\prod_p X_p$, and therefore compact.

As P is directed, the sets $X_{<q}$ have the finite intersection property, as long as the X_p are non-empty. Then $X = \bigcap_q X_{<q}$ is also non-empty:

Lemma 8.8.1. *$X = \varprojlim (\, X_p \mid p \in P\,)$ is a compact Hausdorff space. It is non-empty if $X_p \neq \emptyset$ for all $p \in P$.* □

G, V, E, Ω *$P(G)$, P* *G/p* *dummy vertices*

Given a graph $G = (V, E, \Omega)$, consider as $P = P(G)$ the set of all finite partitions of V with only finitely many cross-edges. Letting $p \leqslant q$ whenever q refines p makes P into a directed partially ordered set. For each p, let G/p be the finite multigraph on p whose edges are the cross-edges of p.[15] The vertices of G/p that are non-singleton partition classes are its *dummy vertices*. The other vertices of G/p, those of the form $\{v\}$, we consider to be vertices of G and refer to them as v.

X_p, f_{qp}

On the compact spaces $X_p := |G/p|$ we have compatible quotient maps $f_{qp}\colon X_q \to X_p$ for $q > p$ which send the vertices of G/q to the vertices of G/p that contain them as subsets; which are the identity on the edges

[15] If the partition classes $U \in p$ are connected in G, then G/p is the minor of G obtained by contracting them. But we do not require them to be connected.